

光纤光栅

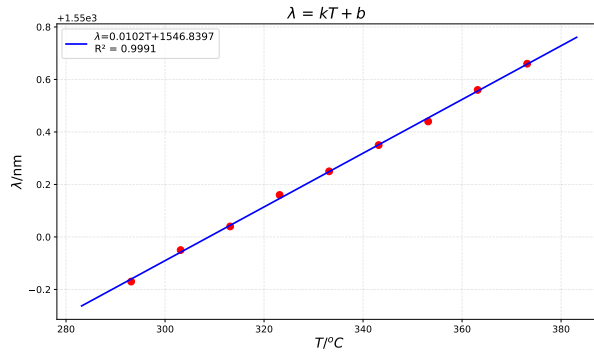
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1 波长随温度的变化



我们知道

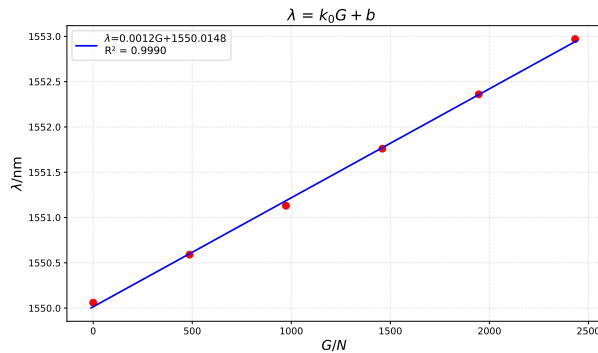
$$\frac{\Delta\lambda_B}{\lambda_B} = \alpha\Delta T \rightarrow \frac{\Delta\lambda_B}{\Delta T} = \alpha\lambda_B = 0.0102nm/K \quad (1)$$

$$\alpha = \frac{0.0102nm/K}{\bar{\lambda}_B} = 6.601090577570692 \times 10^{-6}1/K \quad (2)$$

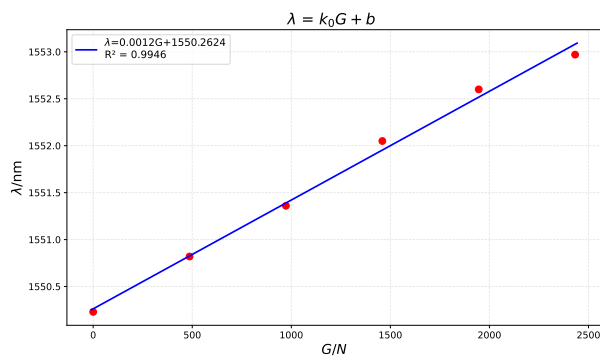
2 波长随拉力的变化

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - P_e)\epsilon = 0.22k\Delta G \quad (3)$$

2.1 拉力增大的过程



2.2 拉力减小的过程



$$\frac{\Delta\lambda_B}{\Delta G} = 0.22k\lambda_B = 0.0012nm/N \quad (4)$$

$$k = \frac{0.0012nm/N}{0.22\lambda_B} = 3.515270383368553 \times 10^{-6}1/N \quad (5)$$

我们回到波长和质量的关系, 画出关系图

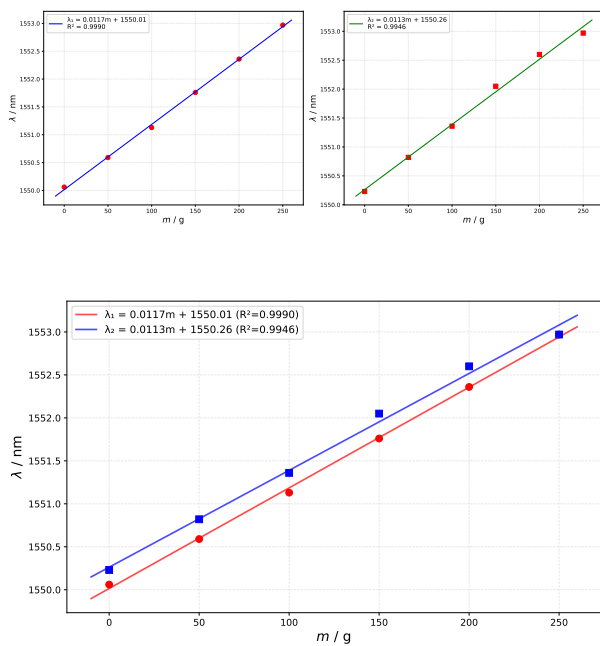


图 1: 红色的线是拉力增大的曲线, 蓝色的线是拉力减小的曲线